

C. To Become Max

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given an array of integers a of length n .

In one operation you:

- Choose an index i such that $1 \leq i \leq n - 1$ and $a_i \leq a_{i+1}$.
- Increase a_i by 1.

Find the maximum possible value of $\max(a_1, a_2, \dots, a_n)$ that you can get after performing this operation at most k times.

Input

Each test contains multiple test cases. The first line of input contains a single integer t ($1 \leq t \leq 100$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains two integers n and k ($2 \leq n \leq 1000$, $1 \leq k \leq 10^8$) — the length of the array a and the maximum number of operations that can be performed.

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^8$) — the elements of the array a .

It is guaranteed that the sum of n over all test cases does not exceed 1000.

Output

For each test case output a single integer — the maximum possible maximum of the array after performing at most k operations.

Example

input	output
6	4
3 4	7
1 3 3	179
5 6	5
1 3 4 5 1	7
4 13	6
1 1 3 179	
5 3	
4 3 2 2 2	
5 6	
6 5 4 1 5	
2 17	
3 5	

Note

In the first test case, one possible optimal sequence of operations is:

$[1, 3, 3] \rightarrow [2, 3, 3] \rightarrow [2, 4, 3] \rightarrow [3, 4, 3] \rightarrow [4, 4, 3]$.

In the second test case, one possible optimal sequence of operations is:

$[1, 3, 4, 5, 1] \rightarrow [1, 4, 4, 5, 1] \rightarrow [1, 5, 4, 5, 1] \rightarrow [1, 5, 5, 5, 1] \rightarrow [1, 5, 6, 5, 1] \rightarrow [1, 6, 6, 5, 1] \rightarrow [1, 7, 6, 5, 1]$.